# Waseem Raza, PhD (Electrical and Computer Engineering)

**Electrical Engineering Technology Educator | Lecture & Laboratory Instruction**

Computer Programming | Communication Systems | Wireless Technology | Applied Engineering

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LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio


## Academic Profile

PhD in Electrical and Computer Engineering with university-level teaching experience in Electrical Engineering Technology at the University of Lahore, Pakistan. Previously taught Computer Programming, Communication Systems, and Wireless Communication in its Department of Technology; supported communications lectures, laboratory experiments, assignment preparation, and assessment at UET Taxila. Experience mentoring students, preparing applied instructional materials, and translating theory into reproducible simulations and practical engineering exercises. Industrial experience with AT&T Labs, Futurewei, and AT&T adds context for telecommunications and technology instruction.

### Research Interests

Applied Electrical Engineering Technology; Communication and Wireless Systems; Engineering Laboratory Instruction; Computer Programming and Measurement Analysis; Industry-Relevant Networked Systems

### Doctoral Research Focus

PhD research in AI/ML for wireless networks, integrating network measurement, communication-system simulation, data-driven modeling and practical reliability analysis.

## Education

**PhD, Electrical and Computer Engineering** *09/2019 - 07/2025*

University of Oklahoma, Norman, Oklahoma, USA

**MSc, Telecommunication Engineering** *08/2014 - 08/2016*

University of Engineering and Technology, Taxila, Pakistan

**BSc, Telecommunication Engineering** *11/2010 - 06/2014*

University of Engineering and Technology, Taxila, Pakistan

### Relevant Graduate Coursework

LTE Networks and Telecommunications; Emerging 5G Systems; Advanced ML and Computer Vision; Wireless Testbed Design; Numerical Analysis; Information Theory; Digital Signal Processing

## Teaching, Research & Professional Experience

### Teaching & Instruction

**Lecturer, Department of Technology** **12/2017 - 09/2019**

*University of Lahore, Pakistan* *Lahore, Pakistan*

- Taught programming, communication systems, and wireless communication with practical examples.
- Advised senior-design students on projects, academic plans, and early careers.
- Aligned course materials with learning outcomes and accreditation standards.
- Supported accreditation, exams, technical workshops, and student activities.

**Teaching areas:** Computer Programming (previously taught); Communication Systems (previously taught); Wireless Communication (previously taught); Communication Systems Laboratories (instructional support); Engineering Computing and Data Analysis (technical preparation); Related Electrical Engineering Technology Laboratories (subject to assignment)

**Student mentoring and research development**

- Advised undergraduate technology student projects and guided programming and communications applications.
- Supported student laboratory experiments, assessment, reports, and debugging of practical implementations.
- Mentored undergraduate and graduate research students in simulations, experiment design, documentation and presentations.

### University Research Appointments

**Graduate Research Assistant** **09/2019 - 07/2025**

*AI4Networks, University of Oklahoma* *Tulsa, OK*

- Built 3GPP-compliant Python simulation tools and an AI-enabled 5G research testbed.
- Built GAN, transfer-learning, autoencoder, and reinforcement-learning models for wireless data.
- Led 100+ experiments testing wireless ML under domain shift and limited labeled data.
- Developed self-healing and outage-management methods, improving resilience by 30%.
- Published wireless AI research and mentored junior researchers in experimental design.

**Research Associate, ComSens Lab** **06/2016 - 12/2017**

*COMSATS University Islamabad, Pakistan* *Islamabad, Pakistan*

- Modeled wireless communication systems through simulation, analysis, and technical reports.
- Mentored Bachelor's and Master's theses in experiment design and research presentation.
- Reviewed student reports and manuscripts for rigor and publication readiness.
- Supported proposals, literature reviews, lab seminars, and research coordination.

**Full-Time Research Scholar, Communication Systems Lab** **08/2014 - 02/2016**
*University of Engineering and Technology, Taxila, Pakistan* *Taxila, Pakistan*

- Researched wireless communication systems while completing an MSc in Telecommunications.
- Supported communications courses, labs, assignments, grading, and examinations.
- Led wireless communication labs and prepared technical manuals and experiments.
- Developed course and student-project materials for practical wireless engineering.

### Industry Research & Engineering

**Machine Learning Engineer** **02/2025 - Present**
*AT&T, contractor via Innovatix Inc.* *Bedminster, NJ*

- Applied domain-adaptive ML to telecom service assurance and model validation.
- Designed cross-domain evaluation and error-analysis methods for operational ML.
- Worked with engineering and public-safety teams on reliable network assurance.

**Data Scientist and AI/ML Engineer** **09/2024 - 12/2024**
*Impac Exploration Services* *Weatherford, OK*

- Built LLM-enabled analytics to interpret geochemical well data for decision support.
- Built Python pipelines in Azure Databricks, Functions, and Synapse Analytics.
- Automated exploration-data ETL, cutting preparation time by 40%.
- Used data-validation and quality controls to improve output reliability by 20%.
- Turned geologists' technical needs into structured datasets and useful analytics.

**Advanced-Degrees Intern** **06/2024 - 08/2024**
*AT&T Labs* *Austin, TX*

- Analyzed 500K+ LTE/5G-NR logs with ML, reducing diagnostic time by 20%.
- Improved 5G NR-NTN cell selection by 15% through data-driven validation.
- Presented operational data insights to next-generation wireless engineering teams.

**Advanced Wireless Technology Development Intern** **09/2022 - 01/2023**
*Futurewei Technologies* *Schaumburg, IL*

- Improved wireless CSI compression by 25% with transfer learning and autoencoders.
- Ran 50+ experiments on wireless ML robustness under changing data conditions.
- Translated research results into recommendations for future wireless networks.

**Advanced-Degrees Intern** **06/2022 - 08/2022**
*AT&T Labs* *Bedminster, NJ*

- Improved 5G standalone user-timing precision by 20% with ML and statistics.
- Analyzed user traces with SQL and presented results to engineering leadership.
- Turned operational traces into indicators for network diagnosis and optimization.

## Research Contributions

### Publications

**AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks** *Computer Communications* 2025 | DOI

**On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks** *Sensors 2025* 2025 | DOI

**An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin** *IEEE PIMRC* 2024 | DOI

**Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity** *IEEE VTC-Fall 2024* 2024 | Link

### Research Projects & Platforms

**SyntheticNET: 3GPP-Compliant Network Simulation** *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

*Tools & Methods:* Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

**Propagation Models & Digital Twins** *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

*Tools & Methods:* Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

**TurboRAN Experimental Platform** *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

*Tools & Methods:* 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

## Research Methods & Technical Expertise

**Undergraduate Technology Teaching:** Computer Programming, Communication Systems and Wireless Communication taught within the University of Lahore Department of Technology; assignments, assessment and student-project guidance

**Engineering Labs & Instruction:** Communications laboratory support, experiment preparation, simulation-based demonstrations, student guidance, lab reports, instructional material and evaluation

**Applied Electrical & Wireless Systems:** Analog/digital communication concepts, RF/wireless fundamentals, signal measurements, LTE/4G and 5G systems, propagation, performance, systems testing

**Programming & Engineering Tools:** Python, C/C++, MATLAB, NumPy, Pandas, data visualization, simulation, Git, Linux and technical documentation

**Instructional Practice:** Clear learning objectives, worked examples, competency-based exercises, outcome evaluation, accessible feedback, attendance/assessment practices and student support

## Academic Service & Recognition

### Academic Service

- Contributed to course material, examinations, curriculum alignment and student academic guidance.
- Supported laboratory documentation, technical reviews and student-project assessment.

### Honors & Awards

**William H. Barkow Scholarship** *2024*
University of Oklahoma, Electrical and Computer Engineering

**Clyde L. Farrar Endowment Scholarship** *2024*
University of Oklahoma

**P.H. Robinson Fellowship** *2023*
University of Oklahoma

**FAS-Charles Forgan Scholarship** *2023*
University of Oklahoma

**Nettie Vincent Boggs Graduate Endowment** *2022*
University of Oklahoma

**SHS-Paul D. Massad Scholarship** *2022*
University of Oklahoma

**Cleo Cross Memorial Scholarship** *2021*
University of Oklahoma

**Marjorie Haskell Engineering Scholarship** *2021*
University of Oklahoma

**Fully Funded PhD / Graduate Research Assistantship** *2019*
University of Oklahoma

## Self-Learning

**1. 4G Network Fundamentals** Mines-Télécom/Coursera | Completed | Aug 2026
LTE radio and core networks, cellular architecture, mobility management, and 4G fundamentals.

**2. AI Foundations & Industry Overview for Telecommunication** Coursera | Completed | Jun 2026
AI applications in telecom networks, operations analytics, service assurance, and automation.

**3. Network and Security Optimization in Telecommunication** Coursera | Completed | Jun 2026
Data-driven network optimization, telecom security, reliability analysis, and operational efficiency.

**4. Python for Data Science, AI & Development** IBM/Coursera | Completed | Jul 2026
Python programming, data analysis, APIs, and reproducible data-science workflows.